Architecture Selection – Hyperlocal Courier Dispatch & Tracking Engine

Architectural Choice

We chose **Microservices Architecture** for the Hyperlocal Courier Dispatch & Tracking Engine. The system contains several distinct functions, including courier request management, rider assignment, request handling, and delivery tracking. These functions can be separated into independently managed services.

Reason 1 – Independent Scaling

The system may handle multiple courier requests, rider assignments, and delivery status updates simultaneously. Microservices allow individual services, such as the **Rider Assignment Service** and **Delivery Tracking Service**, to be scaled independently according to their workload.

Reason 2 – Fault Isolation

The system requires **99% monthly availability**. Separating the major functions into independent services provides better fault isolation, since a failure in one service does not necessarily require the entire system to stop operating.

Security Advantage

Each service can expose only the interfaces required by other components. Access controls can therefore be applied at individual service boundaries, helping restrict unauthorized access to courier, rider, and delivery information.

Performance Benefit

The dedicated **Delivery Tracking Service** can process delivery status updates independently. This supports the requirement that status updates should appear within **3 seconds** and allows the tracking functionality to be scaled separately when required.